

For the period from 1926 through 2016, Morningstar data reveals that the S&P 500 enjoyed an average (arithmetic) return of 12 percent, while intermediate-term government bonds earned 5.3 percent.

When simulating long-term financial plans, we also have to account for volatility and the lack of symmetry in outcomes for positive and negative returns. We calculate the compounded returns over time to account for this volatility. The S&P 500 compounded return fell to 10 percent, while the compounded real return for the less volatile bonds fell only slightly to 5.1 percent. These compounded returns express the growth rate for a portfolio over a number of years, while the larger arithmetic average returns represent the growth rate only for a single year. Compounded returns are smaller than the arithmetic returns because a portfolio loss must be followed by a greater portfolio gain in order to get back to the same starting point. For instance, if a portfolio loses 50 percent of its value, then from this new lower point it must gain 100 percent (not 50 percent) to get back to its initial starting level. At least one popular radio host likes to imply that his listeners' stock portfolios will grow at 12 percent, but this is a misunderstanding on his part. It is the compounded return, not the arithmetic return, that matters for the long-run growth of a portfolio.

But we are still not done. Next, we must remove inflation so the numbers allow for a better understanding of purchasing power growth. Real returns will be less, because they preserve the purchasing power of wealth over time. The real compounded returns fell to 6.9 percent and 2.2 percent, respectively. Though not shown in the exhibit, the real historical arithmetic returns for stocks and bonds were 8.9 percent and 2.4 percent, respectively.

The next step is to consider asset allocation. Though many articles about long-term investing will assume 8 or 12 percent returns, this implicitly suggests that the investor holds 100 percent stocks. That will rarely be the case, especially for retirees. Consider, instead, a retiree with a 50/50 portfolio rebalanced annually. For the historical data, the estimated arithmetic real return from the 50/50 portfolio was 5.6 percent, and the standard deviation for returns was 10.7 percent. The compounded real return was 5.1 percent for a 50/50 portfolio.